O-A-498b
September 7, 1966

SUPERSEDING
Fed. Spec. O-A-498a
October 28, 1957

FEDERAL SPECIFICATION

AMMONIUM DICHROMATE, PHOTOGRAPHIC (AMMONIUM BICHROMATE)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 **Scope.** This specification covers dry ammonium dichromate for photography and lithography.

1.2 **Classification.** This specification covers one grade of ammonium dichromate designated as "photographic grade". (See 6.3.)

2. APPLICABLE DOCUMENTS

2.1 **Standards.** The following standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standard:

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publication.** The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposal, shall apply.

Governmental:

49 CFR 71 to 90—Interstate Commerce Commission Regulations for Transportation of Explosives and Other Dangerous Articles.

(Interstate Commerce Commission Regulations are a part of the Code of Federal Regulations and are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402. Orders for this publication should cite the title and code number as listed above. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 **Material.** The ammonium dichromate shall be in the form of orange to red granular crystals. The material shall not be hard caked in the container when received.

3.2 **Composition.** The ammonium dichro-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

mate shall meet the requirements in table I.

TABLE I. *Composition*

	Percent by weight	Test paragraph
Ammonium dichromate ($(NH4)_2Cr_2O7$), minimum	99.7	4.3.1
Ammonia (NH_3), minimum	13.25	4.3.2
Chloride (Cl) maximum	0.005	4.3.3
Sulfate (SO_4), maximum	0.06	4.3.4

3.3 **Water-insoluble matter.** The water-insoluble matter shall not exceed 0.02 percent when tested as specified in 4.3.5.

3.4 **pH.** The pH shall be not less than 3.2 when tested as specified in 4.3.6.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for inspection and acceptance.** Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material submitted for delivery at one time.

4.2.1 *Inspection of materials and components.* In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.2.2 *Inspection of the end item.*

4.2.2.1 *Examination of the end item.* Examination of the end item shall be made in accordance with the classification of defects, inspection levels, and acceptable quality levels (AQLs) set forth following. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of the end item chemical for examination under 4.2.2.2 through 4.2.2.5, and shall be expressed in units of shipping containers for examination under 4.2.2.6.

4.2.2.2 *Examination of the end item container for defects in appearance, workmanship, closure, and markings.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Container	Not uniform in shape and size, (as applicable).
	Not container specified.
	Not clean—any evidence of dirt, dust or foreign matter.
	Closures (as applicable), not as specified.
	Any evidence of leakage or seepage from container.
	Any hole, tear, puncture, cracks, severe chip, scuff, dent, or other defect in body of container which may affect serviceability.
	Exterior coating (as applicable), not as specified.
Markings, including precautionary markings	Omitted, illegible, incorrect, incomplete, smeared, or not as specified.

4.2.2.3 *Examination of the end item container for defects in appearance of the end item chemical.* The sample unit for this examination shall be one filled primary container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

Examine	Defect
Appearance	Presence of dirt or foreign matter. Form not as specified. Any caking or lumps.

4.2.2.4 *Examination of the end item container for defects in net contents.* The sample unit for this examination shall be one filled primary container. The average net contents shall be not less than the specified or indicated quantity.

4.2.2.5 *Examination of the primary unit container for defects in volume utilized by contents.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Utilized volume	Less than 80 or more than 90 percent.

4.2.2.6 *Examination of preparation for delivery.* An examination shall be made to determine that the packaging, packing, and marking comply with the requirements of section 5. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects. This examination is not applicable when the primary container is the shipping container.

Examine	Defect
Packaging	Not in accordance with contract requirements. Not level specified. Unit package not as specified. Packaging material (as applicable) not as specified.
Packing	Not in accordance with contract requirements. Not level specified. Inadequate application of components such as incomplete strapping, bulged, or distorted containers. Containers not as specified, closures not accomplished by specified or required methods or materials. Liner material (as applicable) omitted, damaged, or not of material specified.
Packing (cont'd)	Any nonconforming component, component missing, damaged, or otherwise defective affecting serviceability.
Weight	Gross or net weight exceeds specified requirements.

Examine	Defect
Count	Number of primary containers per shipping container (as applicable) not in accordance with contract or specification requirements.
Markings	Interior or exterior markings (as applicable), omitted, illegible, incorrect, incomplete, or not in accordance with contract requirements.

4.2.2.7 *Inspection levels and (AQLs) for examination.* The inspection levels for determining the sample size, and the AQLs expressed in defects per 100 units, shall be as follows:

Examination par.	Inspection level	AQL
4.2.2.2	I	2.5
4.2.2.3	S-2	1.5
4.2.2.4	S-2	1.5
4.2.2.5	S-2	4.0
4.2.2.6	S-2	4.0

4.2.3 *Testing of the end item.* The end item shall be tested for the characteristics specified in the specification. One pound of the material shall be drawn from each of 5 shipping containers and composited. The test sample shall consist of 2 pounds of the composite. Duplicate determinations shall be performed on the composite with no evidence of failure to meet the specified requirements. When a numerical requirement is specified for a quality characteristic, the average of two determinations shall comply with the requirement. When attribute requirement is specified, both determinations shall comply.

4.3 Test procedures.

4.3.1 *Ammonium dichromate content.* Accurately weigh about 5 grams (g.) of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

sample. Dissolve in distilled water and dilute to one liter. Mix well. Transfer a 25-ml. aliquot of the solution to a glass-stoppered flask and dilute with 100 ml. of distilled water. Add 5 ml. of reagent grade hydrochloric acid and 3 ± 0.1 g. of reagent grade potassium iodide. Allow to stand in the dark for 15 minutes. Dilute with another 100 ml. of distilled water, and titrate the liberated iodine with 0.1*N* sodium thiosulfate solution, adding 5 ml. of starch indicator solution (see 4.3.1.1) when the iodine color is nearly discharged. Correct the thiosulfate consumed in a blank test made with the same quantities of the same reagents. (Factor: 1 ml. of 0.1*N* sodium thiosulfate solution is equivalent to 0.0042 g. ammonium dichromate.)

4.3.1.1 *Starch indicator solution.* Stir up about 5 g. of soluble starch in 100 ml. of a 1 percent solution of salicylic acid. Add 300 to 400 ml. of boiling water, and boil until the starch dissolves. Dilute to 1 liter.

4.3.2 *Ammonia content.* Accurately weigh 2.0 to 2.5 g. of the sample. Dissolve in 150 ml. of distilled water in a Kjeldahl flask; connect by means of a trap with a well-cooled condenser. The tip of the condenser tube should extend beneath the surface of exactly 50 ml. of 0.5*N* sulfuric acid contained in a receiving flask. To the Kjeldahl flask, add 20 ml. of a 10 percent sodium hydroxide solution. Quickly connect the trap and distill, collecting about 120 ml. of distillate. Titrate the excess of standard acid in the distillate with 0.5*N* sodium hydroxide, using methyl red indicator solution. From the total 0.5*N* sulfuric acid used minus the excess, calculate the percent ammonia in the sample. (Factor: 1 ml. of 0.5*N* sulfuric acid is equivalent to 0.00852 g. of ammonia.)

4.3.3 *Chloride content.* Dissolve 1 ± 0.01 g. of the sample in 20 ml. of distilled water. Add 10 ml. of reagent grade nitric acid, heat to about 50°C., and add a few drops of 0.1*N* silver nitrate solution. Any opalescence produced shall not exceed that produced with the same quantity of reagents to which has been added 0.05 milligrams (mg.) of chlorine as chloride.

4.3.4 *Sulfate content.* Dissolve 20 ± 0.01 g. of the sample in 100 ml. of distilled water. Filter and add 20 ml. of reagent grade hydrochloric acid and 30 ml. of methanol. Boil to expel aldehydes. Dilute to 300 ml. with distilled water. Heat to boiling and slowly add 15 ml. of a 10 percent barium chloride solution. Evaporate almost to dryness on a steam bath and take up in 5 ml. of reagent grade hydrochloric acid and 100 ml. of distilled water. Filter through a tared filter crucible. Wash thoroughly, ignite, cool, and weigh. From the weight of barium sulfate obtained, calculate the percent sulfate (SO_4). (Factor: weight of $BaSO_4$ X 0.4115 weight of sulfate (SO_4).)

4.3.5 *Water-insoluble matter.* Accurately weigh about 10 g. of sample. Dissolve in 100 ml. of distilled water. Add 1 ml. of ammonium hydroxide and allow to stand on a steam bath for 1 hour. Filter through a tared filter crucible and wash the crucible and residue with warm distilled water until the washings are colorless (about 100 ml.). Dry at 105° to 110°C. to constant weight. Calculate the increase in weight to percent insolubles in the sample.

4.3.6 *pH.* Dissolve 200 ± 1 g. of the sample in freshly boiled and cooled distilled water. Dilute to one liter and determine the pH by means of a glass electrode pH meter.

5. PREPARATION FOR DELIVERY

5.1 **Packaging and packing.** Levels of packaging and packing shall be in accordance with the Code of Regulations, Title 49, of the Interstate Commerce Commission and as specified in the contract or order (see 6.2).

5.2 **Marking.**

5.2.1 *Special labeling.* Each container shall be labeled in accordance with the Code of Regulations, Title 49, of the Interstate Com-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.

merce Commission. In addition, each individual container and each shipping container shall carry a "WARNING" label, printed in bold type and securely affixed to the container, worded essentially as follows:

"WARNING—FLAMMABLE
HARMFUL IF SWALLOWED
HARMFUL DUST
MAY CAUSE RASH OR
EXTERNAL ULCERS

Contains Ammonium Dichromate.
Keep away from heat, sparks, and open flame.
Keep container closed.
Avoid contact with skin and eyes.
Avoid breathing dust or solution spray.
In case of contact, immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.
If swallowed, give milk or water and call a physician.
Wash clothing daily.
Use fresh clothing daily.
Keep out of the reach of children."

5.2.2 *Civil agencies.* In addition to the marking specified in 5.2.1, shipments shall be marked in accordance with Fed. Std. No. 123.

5.2.3 *Military agencies.* In addition to the marking specified in 5.2.1 and any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Use.** The ammonium dichromate covered by this specification is intended for use in photographic and lithographic processes.

6.2 **Ordering data.** Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Number, title, and date of this specification.
(*b*) Side and kind of container required (see 5.1).
(*c*) Levels of packaging and packing required (see 5.1).

6.3 **Photographic grade.** The term "photographic grade" refers to chemicals in which impurities known to be photographically harmful are limited to a safe quantity, while inert impurities are restricted to amounts not reducing the required assay strength.

CUSTODIANS:
Army—EL
Navy—WP
Air Force—85

User Activities:
Army—MO
Navy—MC

MILITARY INTERESTS:
Preparing Activity:
Navy—WP

Review/user information is current as of the date of this document. For further coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:14Z
Check the source to verify that this is the current version before use.